

Stat 433 HW 6

1. A and B subsets of sample space S, $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cup B) = 0.5$

a) Are $A \cap B = \emptyset$ not disjoint because $P(A \cup B)$ would be $P(A) + P(B)$, $0.4 + 0.3 \neq 0.5$

$$b) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) + P(B) - P(A \cup B)}{P(B)} = \frac{0.4 + 0.3 - 0.5}{0.3} = \frac{0.2}{0.3} = \boxed{0.6\bar{6}}$$

2. A and B events in sample space S, $P(A) = 0.25$, $P(B) = 0.4$, $P(A \cup B) = 0.6$

a) $P(A^c) = \boxed{0.75} = 1 - P(A)$

$P(B^c) = 0.6$

b) $P(A \cap B) = 0.25 + 0.4 - 0.6 = \boxed{0.05} = P(A) + P(B) - P(A \cup B)$

c) $P(A \cap B^c) = P(A) - P(A \cap B) = 0.25 - 0.05 = \boxed{0.2}$

d) $P(A^c \cap B^c) = 1 - 0.6 = \boxed{0.4} = 1 - P(A \cup B)$

e) $P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{0.05}{0.25} = \boxed{0.2}$

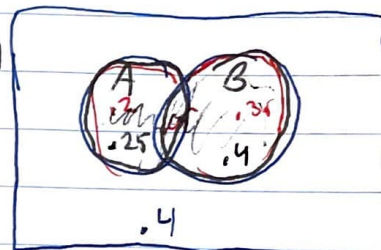
f) are A and B independent? Explain.

Are $P(A \cap B) = P(B)P(A)$ or $P(B|A) = P(B)$

$0.05 \neq 0.1$ and

$0.2 \neq 0.4$, neither true, they are

not independent



3. Randomly chosen automobile needs oil change $P(O) = 0.25$, Needs new oil filter $P(F) = 0.4$

Needs both: $P(O \cap F) = 0.14$

$P(O) = 0.25$, $P(F) = 0.4$, $P(O \cap F) = 0.14$

a) $P(O \cup F)$

$= P(O) + P(F) - P(O \cap F) = 0.25 + 0.4 - 0.14 = \boxed{0.51}$

b) $P(F|O) = \frac{P(F \cap O)}{P(O)} = \frac{0.14}{0.25} = \boxed{0.56}$

c) $P(O|F) = \frac{P(O \cap F)}{P(F)} = \frac{0.14}{0.4} = \boxed{0.35}$

d) Are O and F independent? Explain.

Are $P(O \cap F) = P(O)P(F)$ or $P(O|F) = P(O)$

$0.14 \neq 0.1$

$0.35 \neq 0.25$

Not independent